



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A' Grade

Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 5	
Name of Student	Raghav Dhoot
PRN	23070122170
Batch	2023 - 2027
Class	CSE - B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Jenkins Setup & Freestyle Project
Implementation & Theory	<pre>[(base) raghavdhoot@Raghavs-MacBook-Air-2 ~ % docker --version Docker version 29.6.2, build dfc4efb] (base) raghavdhoot@Raghavs-MacBook-Air-2 ~ % docker run -d -p 8080:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home --name myjenkins jenkins/jenkins:lts-jdk17 6f38c682b1c1dd4370fda96f6f6c0904a34dbba594f3a2a00f126f5eaf5453 (base) raghavdhoot@Raghavs-MacBook-Air-2 ~ % docker ps CONTAINER ID IMAGE COMMAND CREATED STATUS 6f38c682b1c1 jenkins/jenkins:lts-jdk17 "/usr/bin/tini -- /u- 10 seconds ago Up 10 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp myjenkins</pre> docker ps (cleanup + fresh run) — Removed the old myjenkins container and its volume to start clean, then ran a fresh Jenkins LTS container via docker run, confirmed with docker ps showing the container up with ports 8080 (web UI) and 50000 (agent communication) mapped. <pre>(base) raghavdhoot@Raghavs-MacBook-Air-2 ~ % docker exec myjenkins cat /var/jenkins_home/ secrets/initialAdminPassword 4f07e0c7804b4b4d6c9daf72b61718d7fb</pre> Initial admin password — Retrieved the auto-generated unlock password from inside the container using docker exec myjenkins cat /var/jenkins_home/secrets/initialAdminPassword.

Getting Started

Create First Admin User

Username
raghav11

Password
.....

Confirm password
.....

Full name
Raghav Dhoot

E-mail address
iamraghavdhoot@gmail.com

Jenkins 2.541.3 [Skip and continue as admin](#) [Save and Continue](#)

Unlock Jenkins — Pasted the password into the "Unlock Jenkins" page at localhost:8080 to proceed with setup.

Install suggested plugins — Selected "Install suggested plugins," which installed the standard plugin set including Git and Pipeline.

Create admin user — Filled in username, password, full name, and email to create the first admin account.

 **Jenkins**

[+ New Item](#)
[Build History](#)

Build Queue

No builds in the queue.

Build Executor Status

0/2

Search

Settings

Help

[Add description](#)

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

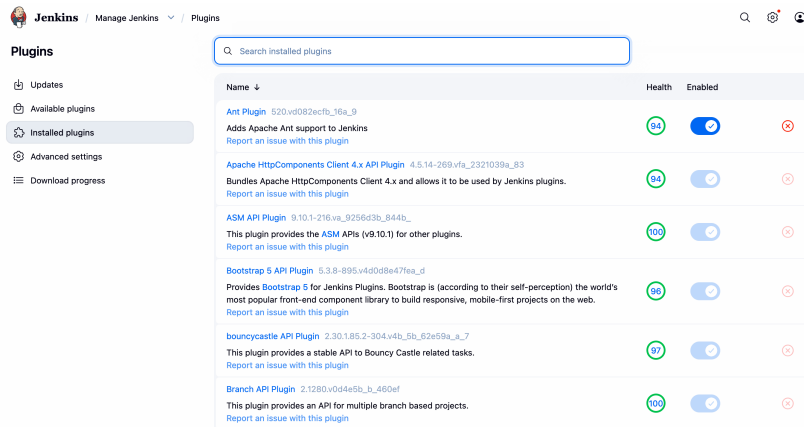
Set up an agent

Configure a cloud

Learn more about distributed builds

Instance configuration — Confirmed the Jenkins URL as http://localhost:8080/ and completed setup, landing on the Jenkins

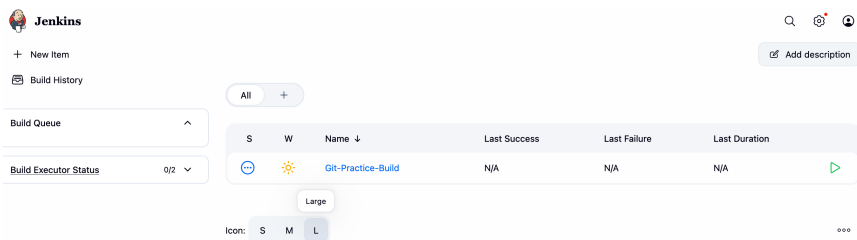
dashboard.



The screenshot shows the Jenkins 'Manage Jenkins' → 'Plugins' page. The 'Installed plugins' tab is selected. A search bar at the top contains 'Search installed plugins'. Below it, a table lists installed plugins with columns for Name, Health, and Enabled status.

Name ↓	Health	Enabled
Ant Plugin 520.vd082ecfb_16a_9 Adds Apache Ant support to Jenkins Report an issue with this plugin	94	On
Apache HttpComponents Client 4.x API Plugin 4.5.14-269.vfa_2321039a_83 Bundles Apache HttpComponents Client 4.x and allows it to be used by Jenkins plugins. Report an issue with this plugin	94	On
ASM API Plugin 9.10.1-216.va_9256d3b_844b_ This plugin provides the ASM APIs (v9.10.1) for other plugins. Report an issue with this plugin	100	On
Bootstrap 5 API Plugin 5.3.8-895.v4d0d8e476ea_d Provides Bootstrap 5 for Jenkins Plugins. Bootstrap is (according to their self-perception) the world's most popular front-end component library to build responsive, mobile-first projects on the web. Report an issue with this plugin	96	On
bouncycastle API Plugin 2.30.1.85.2-304.v4b_5b_62e59a_a_7 This plugin provides a stable API to Bouncy Castle related tasks. Report an issue with this plugin	97	On
Branch API Plugin 2.1280.v0d4e5b_b_460ef This plugin provides an API for multiple branch based projects. Report an issue with this plugin	100	On

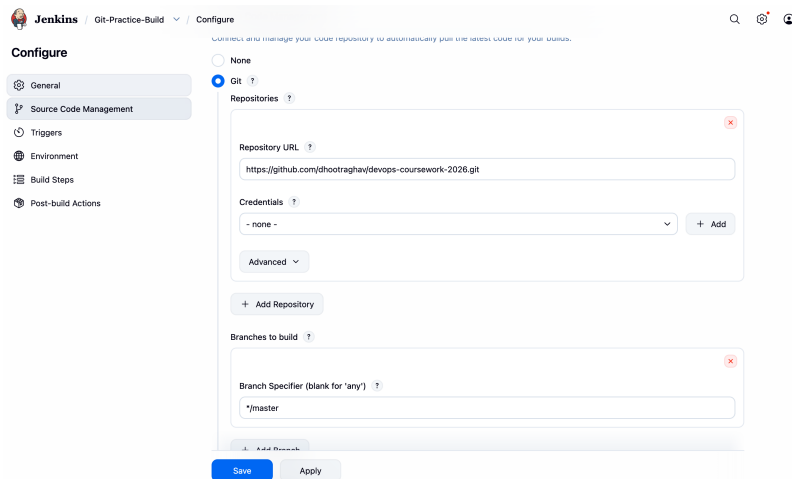
Verify Git plugin — Checked Manage Jenkins → Plugins → Installed, confirming the Git plugin was present from the suggested install.



The screenshot shows the Jenkins 'Build Queue' page. The 'Build Queue' tab is selected. It displays a table of builds with columns for S, W, Name, Last Success, Last Failure, and Last Duration. A single build is listed: 'Git-Practice-Build'.

S	W	Name ↓	Last Success	Last Failure	Last Duration
...	...	Git-Practice-Build	N/A	N/A	N/A

Create Freestyle project — Created a new item named Git-Practice-Build, selected "Freestyle project" as the type.



The screenshot shows the Jenkins 'Configure' page for the 'Git-Practice-Build' project. The 'Source Code Management' tab is selected. The configuration is set to 'Git'. The 'Repository URL' is 'https://github.com/dhootraghav/devops-coursework-2026.git'. The 'Credentials' dropdown is set to '- none -'. The 'Branches to build' section is expanded, showing 'Branch Specifier (blank for 'any')' set to '*/master'.

Configure Source Code Management — Set SCM to Git and pointed it at the devops-coursework repository URL, using the default branch specifier.

Jenkins | Git-Practice-Build | Configure

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell

Command

```
echo "Contents of README.md"
cat README.md

echo ""
echo "Hello Jenkins!"
```

Post-build Actions

Define what happens after a build completes, like sending notifications, archiving artifacts, or triggering other jobs.

Save **Apply**

Add build step — Added an "Execute shell" build step with commands to print job/build info, list workspace files, and output the contents of [README.md](#).

Jenkins | Git-Practice-Build | #3

Console

Started by user Raghav Dhoot

Running as SYSTEM

Building in workspace /var/jenkins_home/workspace/Git-Practice-Build

The recommended git tool is: NONE

No credentials specified

```
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/Git-Practice-Build/.git # timeout=10
> git config remote.origin.url https://github.com/dhootraghav/devops-coursework-2026.git # timeout=10
Fetching upstream changes from https://github.com/dhootraghav/devops-coursework-2026.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/dhootraghav/devops-coursework-2026.git +refs/heads/
+refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 5fc7edfd7c7f74be3bfcde8b3b0d78c3b41673b (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 5fc7edfd7c7f74be3bfcde8b3b0d78c3b41673b # timeout=10
Commit message: "Add files via upload"
First time build. Skipping changelog.
[Git-Practice-Build] $ /bin/sh -xe /tmp/jenkins14985395536599553893.sh
+ echo Listing files checked out from Git:
Listing files checked out from Git:
+ ls -la
total 76
drwxr-xr-x 18 jenkins jenkins 4096 Sep 2 21:16 .
drwxr-xr-x 3 jenkins jenkins 4096 Sep 2 21:13 ..
drwxr-xr-x 8 jenkins jenkins 4096 Sep 2 21:16 .git
-rw-r--r-- 1 jenkins jenkins 0 Sep 2 21:16 .gitkeep
drwxr-xr-x 2 jenkins jenkins 4096 Sep 2 21:16 Assignment - 01
drwxr-xr-x 2 jenkins jenkins 4096 Sep 2 21:16 Assignment - 02
```

Run and verify build — Triggered "Build Now," opened Console Output, and confirmed the build checked out the repo, ran the shell commands, printed README.md contents, and finished with Finished: SUCCESS.